

Puzzles & Synthesis

Name: _____ Date: _____

Patience rule: For knights-and-knaves problems, try each case ("Assume A is a knight...") and run it to the end. If it breaks, record *why* before starting the next case. A correct answer with no case analysis earns partial credit at best.

Deduction problems — Puzzles

D6.1 DEDUCTION

Four people — A, B, C, D — are each a knight (always truthful) or a knave (always lies). You hear:

- **A:** "B and C are the same type."
 - **B:** "D is a knave."
 - **C:** "A is a knave if and only if D is a knight."
 - **D:** "B is a knight and C is a knave."
1. There are 16 possible type assignments. Use structured case analysis: assume A is knight, derive everything that follows; if contradiction, discard. Repeat for A knave.
 2. Within each case, do not stop at the first guess — write whether B, C, D must be knight or knave and verify all four statements.
 3. State the unique solution (or prove none exists).

FULL CASE ANALYSIS + VERIFIED SOLUTION

D6.2 DEDUCTION

Three houses in a row (left, middle, right). Each house has a different pet (cat, dog, fish) and a different color door (red, green, blue). Clues:

1. The cat is not in the middle house.
 2. The red door is immediately to the left of the house with the dog.
 3. The fish is in the house with the green door.
 4. The house on the right does not have a blue door.
- a. Translate each clue using propositional letters (e.g. C_L = cat in left house). Use as many letters as you need.
 - b. Build a table of all 6 permutations of pets $\times$ check door constraints systematically. For each candidate layout, mark each clue T or F.
 - c. Identify the only layout that satisfies all four clues. Show your elimination work — do not jump to the answer.

SYMBOLIZATION + ELIMINATION TABLE + FINAL LAYOUT

D6.3 DEDUCTION

Five suspects — P, Q, R, S, T — exactly one is guilty. Each makes one statement:

- **P:** “The guilty person is Q or R.”
- **Q:** “If P is guilty, then S is innocent.”
- **R:** “T is guilty if and only if P is innocent.”
- **S:** “It is not true that both Q and T are guilty.”
- **T:** “R is guilty and Q is innocent.”

You are told: **exactly four** statements are true and **exactly one** is false (under the actual guilty person).

1. For each suspect $G \in \{P, Q, R, S, T\}$, assume G is guilty and compute T/F for all five statements. Use a 5-row table.
2. Which guilty person is the only one that yields “four true, one false”?
3. Verify: for your answer, write one sentence per statement explaining why it is T or F.

5-ROW ASSUMPTION TABLE + VERIFICATION SENTENCES

D6.4 DEDUCTION

Course synthesis — one long problem, all tools allowed:

Define: p = “The server is online,” q = “The password is correct,” r = “The user has admin rights,” s = “Access is granted.”

System rules:

- R1: Access is granted if and only if the server is online and (the password is correct or the user has admin rights).
- R2: If the password is not correct, then the user does not have admin rights. (Contrapositive form also acceptable.)
- R3: The server is not online unless maintenance mode is off. (Hint: rewrite “A unless B” as $\neg B \rightarrow A$ or equivalent.)

Given: maintenance mode is off, the password is not correct, access is granted.

- a. Symbolize R1–R3. Introduce a letter for “maintenance mode is off” if needed.
- b. Build an 8- or 16-row table (your choice — explain why) and find every row consistent with all givens and rules.
- c. What must be true about p, q, r, s ? Prove any claim you make.
- d. Write a 5-sentence “incident report” in plain English explaining what happened, citing which rules you used.

SYMBOLS + TABLE + FORCED TRUTHS + INCIDENT REPORT